

App Dev Project Report

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About Me: I am currently pursuing an online diploma at IIT Madras while working full-time in the banking sector. I have 13 years of professional experience in banking. I have a strong interest in learning and applying technology in real-world systems.

AI/LLM usage

The estimated AI/LLM usage in this project is approximately 20–30%, mainly limited to debugging assistance, while the overall design and implementation were done independently.

Description

This project involves developing a web-based Placement Portal that connects students, companies, and administrators. Students can apply for jobs, companies can post and manage job openings, and the admin oversees approvals. The system streamlines the placement process through a webpage platform.

Technologies used

Technology	Purpose
Flask	Core backend web framework used to handle routing and application logic
SQLite	Lightweight database used to store students, companies, jobs, and applications
Jinja2	Template engine used to render dynamic HTML pages
HTML	Used to structure web pages
CSS	Used for styling the application
Bootstrap 5	Used for responsive design and improving UI appearance
Python	Core programming language used for backend development

DB Schema Design

Tables:

1. Students

- student_id (PRIMARY KEY)
- Student_name (TEXT, NOT NULL)
- program (TEXT, NOT NULL)
- email (TEXT, UNIQUE, NOT NULL) – Student email
- password (TEXT, NOT NULL) – Login password

2. Companies

- company_id (TEXT, PRIMARY KEY) – Unique company ID (e.g., C101)
- name (TEXT, NOT NULL) – Company name
- email (TEXT, UNIQUE, NOT NULL) – Company email
- password (TEXT, NOT NULL) – Login password
- status (TEXT) – Approval status (Pending/Approved)

3. Jobs

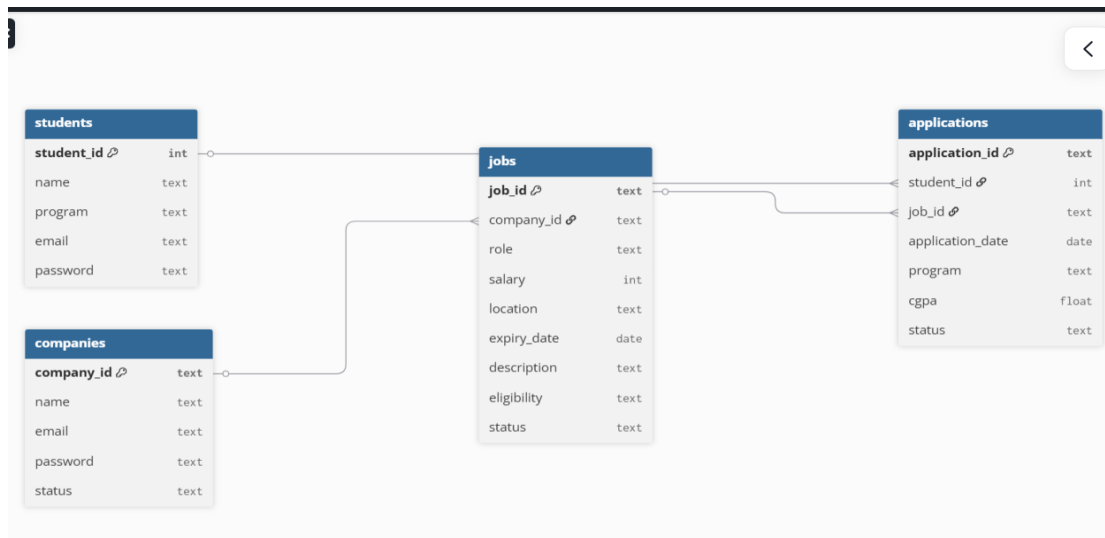
- job_id (TEXT, PRIMARY KEY) – Unique job ID (e.g., J101)
- company_id (TEXT, FOREIGN KEY) – References Companies
- role (TEXT, NOT NULL) – Job role
- salary (TEXT)
- location (TEXT) – Job location
- expiry_date (DATE) – Last date to apply
- description (TEXT) – Job details
- eligibility (TEXT) – Eligibility criteria
- status (TEXT) – Approval status

4. Applications

- application_id (TEXT, PRIMARY KEY) – Unique application ID (e.g., A101)
- student_id (INTEGER, FOREIGN KEY) – References Students
- job_id (TEXT, FOREIGN KEY) – References Jobs
- application_date (DATE) – Date of application
- program (TEXT) – Student program
- cgpa (REAL) – Student CGPA
- status (TEXT) – Application status (Pending/Shortlisted/Selected/Rejected)

Relationships:

- One Student → Many Applications
- One Company → Many Jobs
- One Job → Many Applications



API Design

NA

Architecture and Features

The application follows a simple client-server architecture using Flask as the backend framework. The browser acts as the client, sending requests to the Flask server, which processes the logic and interacts with the SQLite database. The server then returns dynamic HTML pages rendered using Jinja2 templates.

Project Structure:

- app.py – Main Flask application containing routes and logic
- templates/ – HTML files rendered using Jinja2
- static/ – CSS and Bootstrap files for styling
- SQLite Database – Stores students, companies, jobs, and applications

Features Implemented:

- Student registration and login
- Company registration and login
- Admin dashboard for managing students, companies, and jobs
- Job posting by companies
- Job approval by admin
- Students can view and apply for jobs
- Application tracking with status (Pending, Shortlisted, Selected, Rejected)
- Company can view and update application status

Video

https://drive.google.com/file/d/1H1O_KyvmLtUTcBzI9RQMV8kpQ1qEPer-/view?usp=drive_link